Introduction

Drafting is a critical part of building a competitive National Hockey League (NHL) team, as it is the cheapest and easiest way to get top end players long term[1]. Acquiring talent cheaply is vitally important in a salary capped league, as it gives teams more money to be able to sign and trade for other players needed to fit team needs. This is all done with the objective of making and succeeding in the playoffs, as that is the gateway to generating tens if not hundreds of millions of dollars in additional revenue[2,3]. Finding game changing players in the draft is far from a guarantee, however, as over half of players drafted by an NHL team never play a single game in the NHL, which can be a huge missed opportunity for teams. A challenge for teams is that draft eligible players come from a wide variety of different leagues which have varying levels of play. For example, 0.6 points per game could be excellent in the SHL, but not very good in the QMJHL (Figure 1). Not only that, but these potentially future altering draft decisions need to be made in a matter of minutes, as teams only have 3-4 minutes to make each pick on draft day. To help teams make optimal decisions at the draft, I will be developing models for player analysis based on the league they played in prior to the draft. These models will allow team analysts to cut through the noise of league context when determining whether a player's statistics imply that the player has talents that can translate to the NHL. These models will also be designed with speed and ease of use in mind to help teams make decisions on these players quickly.

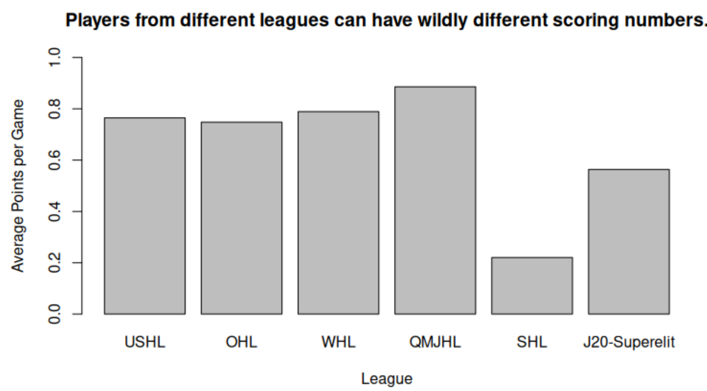

Figure 1: Average points per game by players drafted from each league

Methods

League	Players
USHL	337
OHL	329
WHL	287
QMJHL	188
SHL	141
J20 Superelit	87
LIIGA	68
KHL	60
NCAA	44
USHS-Prep	42
U20 SM-liiga	38
BCHL	31
MHL	25
CZECH	23
AJHL	22
OJHL	18
NLA	12

Table 1: Drafted player count by league

Two datasets were used for this project. The first, scraped from the hockeydb website[4], has all players drafted to the NHL between 2012 and 2021. This dataset also includes the number of games each player has played so far in the NHL, if any. It can take several years for a drafted player to play their first game in the NHL, but most who do do so within four years of their drafting[5]. For this reason, the cutoff year of 2021 was chosen for the analysis. The second dataset, scraped from the eliteprospects website[6], includes player statistics from 17 different hockey leagues, which encompasses the large majority of places where NHL draftees played pre-draft. While there were still some leagues not included in this dataset where players were drafted from, none of these leagues had more than ten such players. These players were simply removed from the analysis, as they did not represent a large enough sample size for those leagues to make sweeping claims about the leagues as a whole. The two datasets were combined, creating a dataset with the prior league and season stats for every remaining player in the draft dataset.

A classification tree was generated on this dataset, predicting whether a player would make the NHL given their stats from the previous league. This method was chosen because the easily interpretable and usable results of the tree lend well to quick decision making, which is critical during a draft. One flaw of this method, however, is that small changes in the data can lead to completely different trees, with different decisions during a draft. For this reason the tree may not be effective as the only tool for determining who to pick, but it still has value for quickly determining if a player's stats are still "good" when given the context of the league they played in. It can also be effective for determining which numbers are impressive regardless of league.

To aid with the classification process, individualized naive Bayes models were trained for each league with over fifty drafted players. 20-fold cross validation was used for the models to reduce training error from the smaller individual datasets. Principal component analysis was also performed on each fold prior to training each model to combat the severe multicollinearity present in the datasets (figure 2). Each model selected the principal components needed to capture 95% of the variance in the dataset. This value was kept quite high as the PCA was needed primarily for multicollinearity, since the datasets did not have very high dimensionality. This overall method of using principal component analysis with a naive Bayes model has also been shown by Macióńczyk et al. to reduce model training time while having an imperceptible drop in accuracy[7].

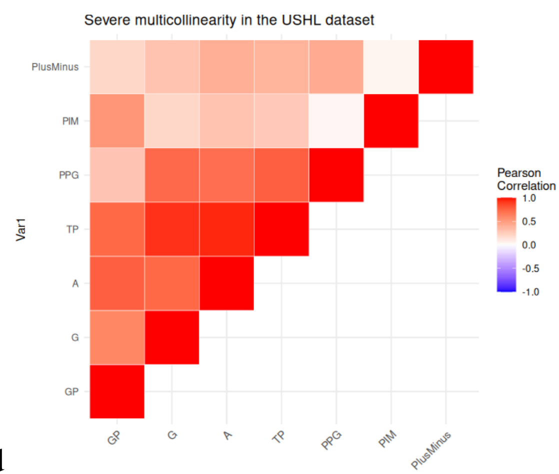

Figure 2: Correlation matrix for the USHL dataset

Results

The full classification tree was trained on the dataset of player pre-draft statistics, classifying players on whether they will play in the NHL. This full tree was very complicated and entirely overfit, so pruning was deemed necessary. Two potential pruned trees were identified. The first had twenty-nine total splits and the lowest cross validation error of all possible trees. The second tree had five splits and had relatively low cross validation error for its small size. Both trees were computed and tested. The second, smaller tree proved to be better both at accuracy (59% vs 54%), ROC AUC (59% vs 57%), and interpretability. This model has a sensitivity of 51.7% and a specificity of 66.2%, meaning the tree is much better at picking out players who will not make the NHL than it is at picking out players who will. This tree (figure 3), therefore, can provide a quick, effective method of eliminating potential players from a team's draft board. This has the potential to be very useful for teams to help quickly narrow down the number of players to consider for a particular draft pick, allowing the team more time to perform deeper analysis on the remaining players. The decile-wise lift chart for this tree (figure 4) shows moderate but not incredible predictive capability. For this reason, this model should not be used as the only determinator of who to draft, but instead its use should be limited to quickly eliminating players who are unlikely to play in the NHL.

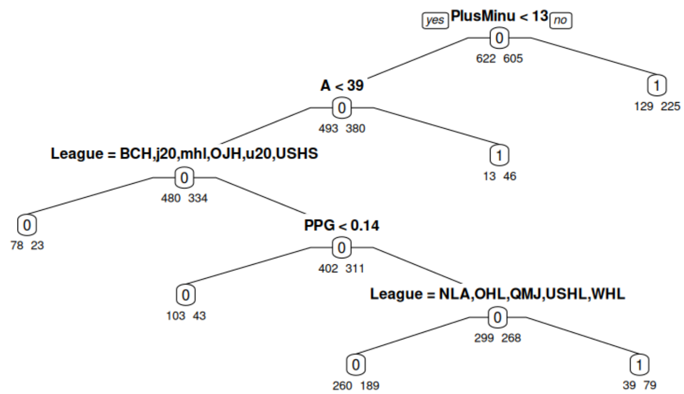

Figure 3: Pruned tree output

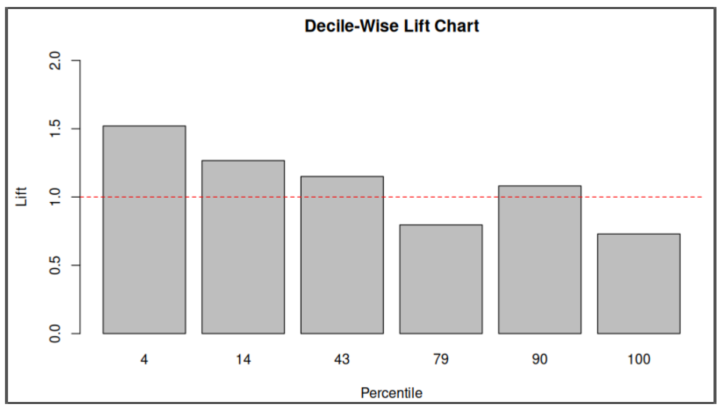

Figure 4: Decile-wise lift chart for the classification tree

According to the resulting classification tree, players who have a plus/minus of at least twelve or at least thirty-nine assists are likely to play in the NHL regardless of the league in which they obtained those stats. It makes sense that plus/minus (a team's goal differential when that player was on the ice) is a useful determinator for future NHL players regardless of league, as the magnitude of players' plus minus tends to remain fairly consistent across different leagues even with differing scoring rates. The number of assists, however, is a bit more surprising as the secondary indicator of playing in the NHL, as that is more heavily reliant on the league scoring environment. Players who have neither a high enough plus/minus or enough assists are only considered likely to play in the NHL if they have at least 0.14 points per game and they play in one of the following leagues: the SHL, the Finnish Liiga, the KHL, NCAA, the Czech Extraliga, or the AJHL. This can imply that those six leagues are either more difficult or lower scoring than the others, as a young player scoring 0.14 points per game is worthy of classification as a future NHLer (given the prior determinations) in just those leagues. This makes sense given that several of these leagues are professional leagues with no age cap, meaning pre-draft players (who are generally younger players) in these leagues are playing with and against older players, so lower numbers are still impressive in these leagues.

Since the classification tree had limited overall classification capabilities, dedicated naive Bayes models were trained for each league with enough sample size. Eight such models were trained. The results of these models are below:

League	USHL	OHL	WHL	QMJHL	SHL	J20-Superelit	Liiga	KHL
Accuracy	0.595	0.639	0.640	0.590	0.610	0.609	0.632	0.644
Sensitivity	0.648	0.638	0.720	0.670	0.652	0.860	0.833	0.710
Specificity	0.546	0.640	0.559	0.511	0.573	0.133	0.406	0.571

Table 2: League model results

These models do provide better overall accuracy than the classification tree model, but not by a lot. What these models do provide, however, is much better sensitivity. All of these league models boasted higher sensitivity than specificity, with the sole exception of the OHL model where the two scores were practically identical. The SHL and J20-Superelit models sport particularly impressive sensitivities, though these do come at the cost of very poor specificity. This means that if one of these models predicts that a player is going to make the NHL, they are very likely correct, but if they predict that a player will not make the NHL, you cannot trust that judgement nearly as much. Unfortunately, due to the use of principal components analysis in the training of these models, they are almost entirely uninterpretable. These models are useful for classifying new data, but they cannot give much insight into the new players beyond whether they will play in the NHL.

Rejected Alternative Approaches

Whether a drafted player makes the NHL is a very simple, understandable, but still reasonably effective way to determine if a draft pick was a “bust”. For this reason, this analysis was phrased as a classification problem, which eliminated the need for any regression techniques. The two types of models used for this analysis were chosen primarily due to their ability to quickly classify new player data, a criterion which eliminated k-nearest neighbors. Interpretability was also valued in this analysis, which is why a simple classification tree was used over a potentially more stable ensemble tree. This interpretability allows the tree to provide additional insight into what statistics can be important for evaluating draft prospects (namely plus/minus and assists).

Conclusion

Drafting is typically the most cost effective way for NHL teams to acquire new players, which allows teams who draft effectively to acquire more players which is critical to making a playoff push. These young, cost-controlled players are extremely valuable assets as they can be developed into or traded for top-end players[8]. Despite the importance of the draft, less than half of drafted players make the NHL. Making poor drafting decisions negatively impacts a team not only competitively, but also monetarily. Teams can generate over $10 million in additional revenue for each playoff game they play, meaning that every season a team spends in the basement of the league trying to acquire strong draft picks is a missed windfall. Drafting becomes difficult, however, when the players available to be drafted come from over a dozen different leagues from all around the globe, with each league having different skill levels and scoring quantities. This analysis helps tackle this problem by revealing the plus/minus statistic as a potentially powerful means of evaluating players regardless of league. Models have developed to determine if a drafted player has what it takes to make the NHL. These models include league specific models which are very effective at selecting players that will most likely be good enough for the big show. Implementation of these findings can help teams acquire more, better assets which can be used to make a better team capable of generating more playoff revenue.

References

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Appendix A: League Abbreviations and Locations

- NHL: National Hockey League (USA and Canada)
- USHL: United States Hockey League (USA)
- OHL: Ontario Hockey League (USA and Canada)
- WHL: Western Hockey League (USA and Canada)
- QMJHL: Quebec Major Junior Hockey League (Canada)
- SHL: Swedish Hockey League (Sweden)
- J20 Superelit (Sweden)
- Liiga (Finland)
- KHL: Kontinental Hockey League (Russia, Belarus, Kazakhstan, and China)
- NCAA: National Collegiate Athletic Association (USA)
- USHS: United States High School (USA)
- U20 SM-liiga: Under 20 Finnish Championship Series (Finland)
- BCHL: British Columbia Hockey League (Canada)
- MHL: Molodyozhnaya Hokkeinaya Liga (Russia, Belarus, and Kazakhstan)
- CZECH (Czechia)
- AJHL: Alberta Junior Hockey League (Canada)
- OJHL: Ontario Junior Hockey League (USA and Canada)
- NLA: National League (Switzerland)

Appendix B: Unused Visuals

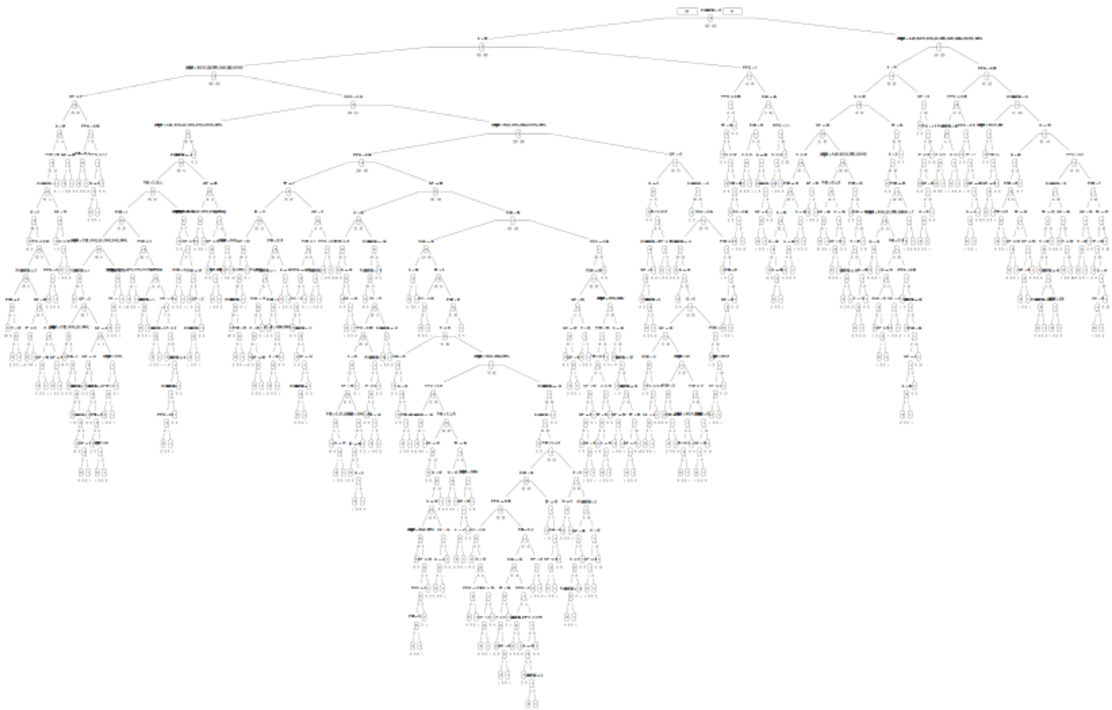

Figure B.1: Full classification tree

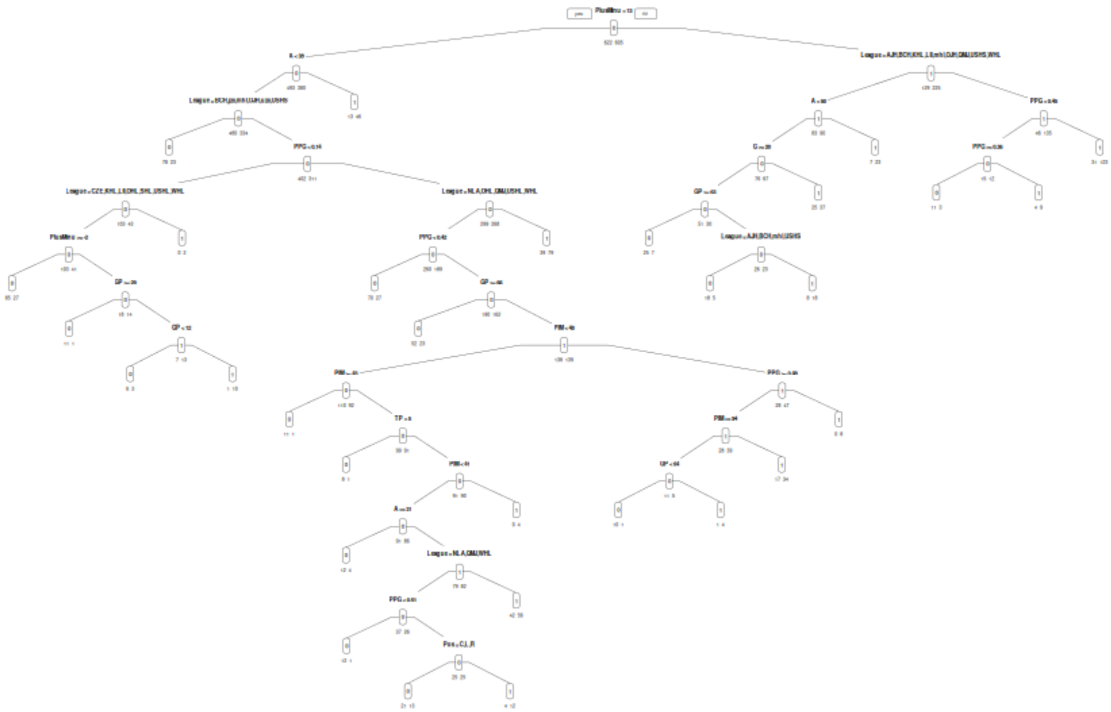

Figure B.2: Rejected pruned classification tree

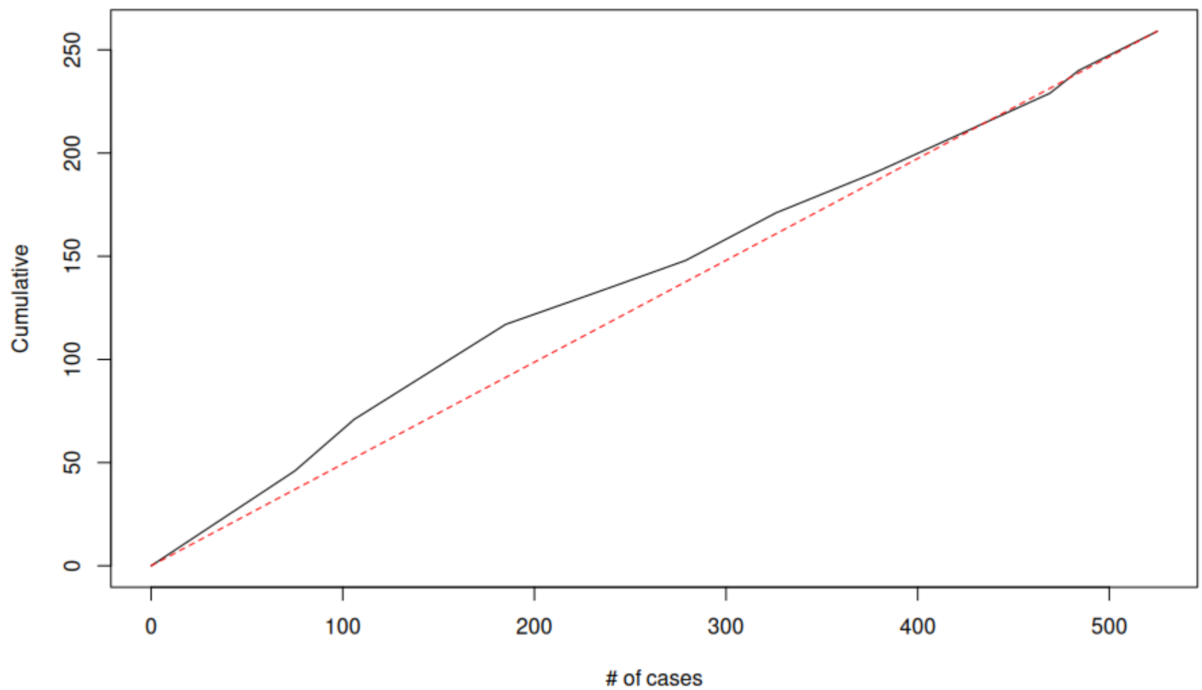

Figure B.3: Cumulative lift chart for the rejected pruned classification tree

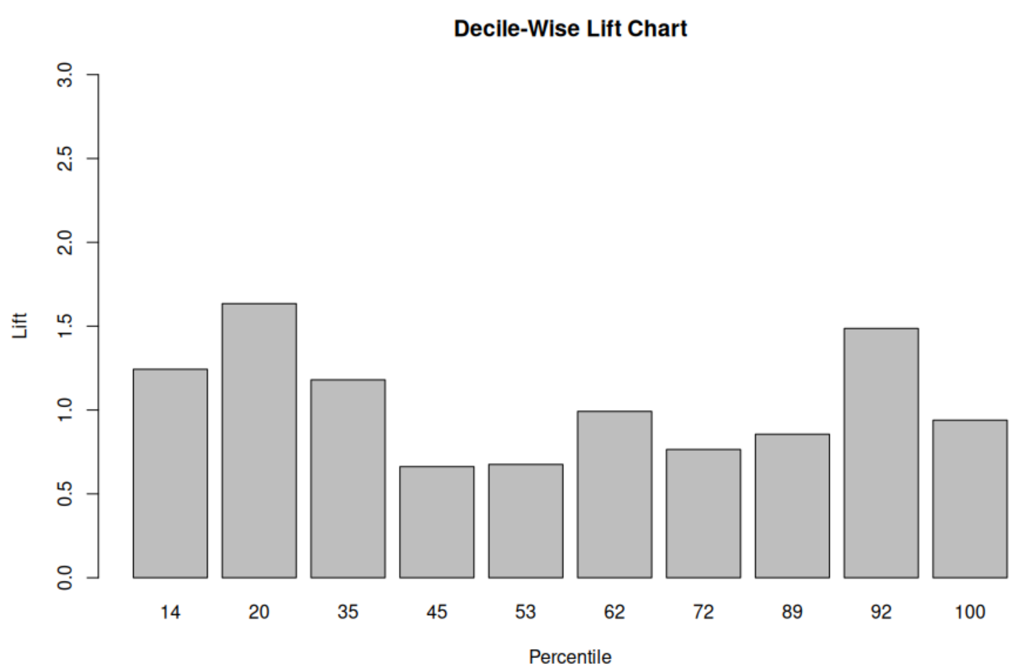

Figure B.4: Decile-wise lift chart for the rejected pruned classification tree

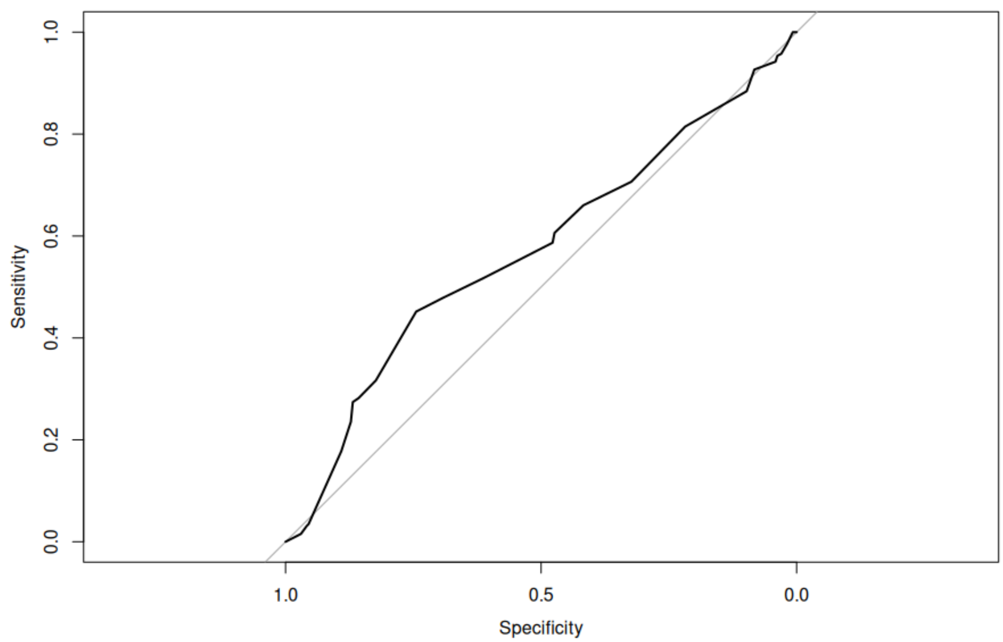

Figure B.5: ROC curve for the rejected pruned classification tree

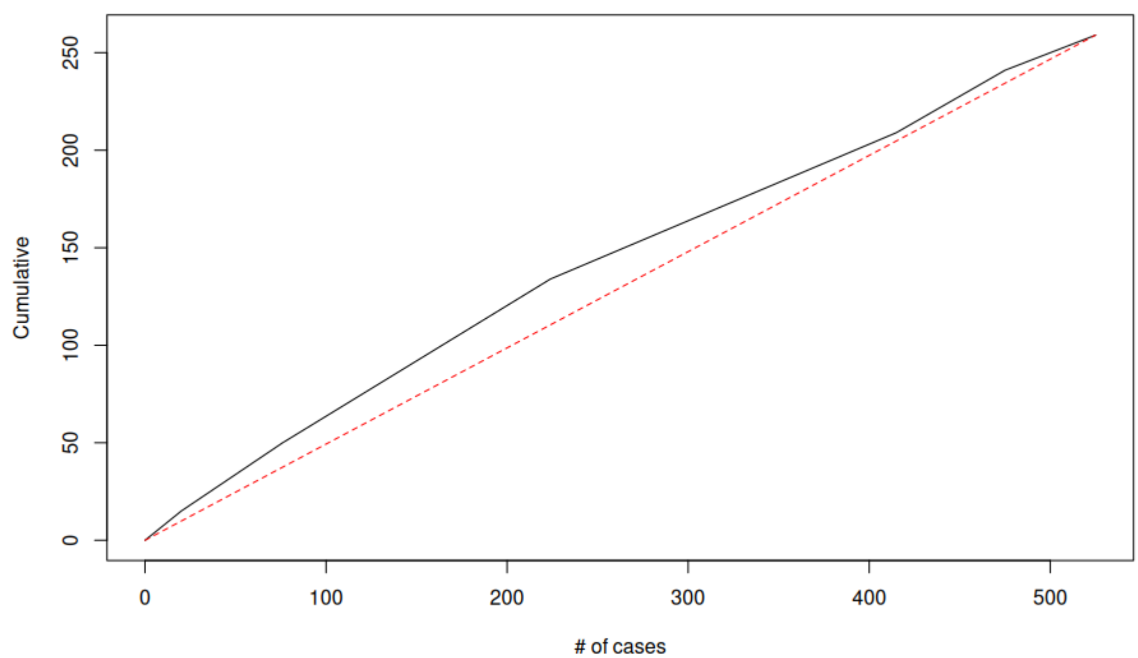

Figure B.6: Cumulative lift chart for the pruned classification tree

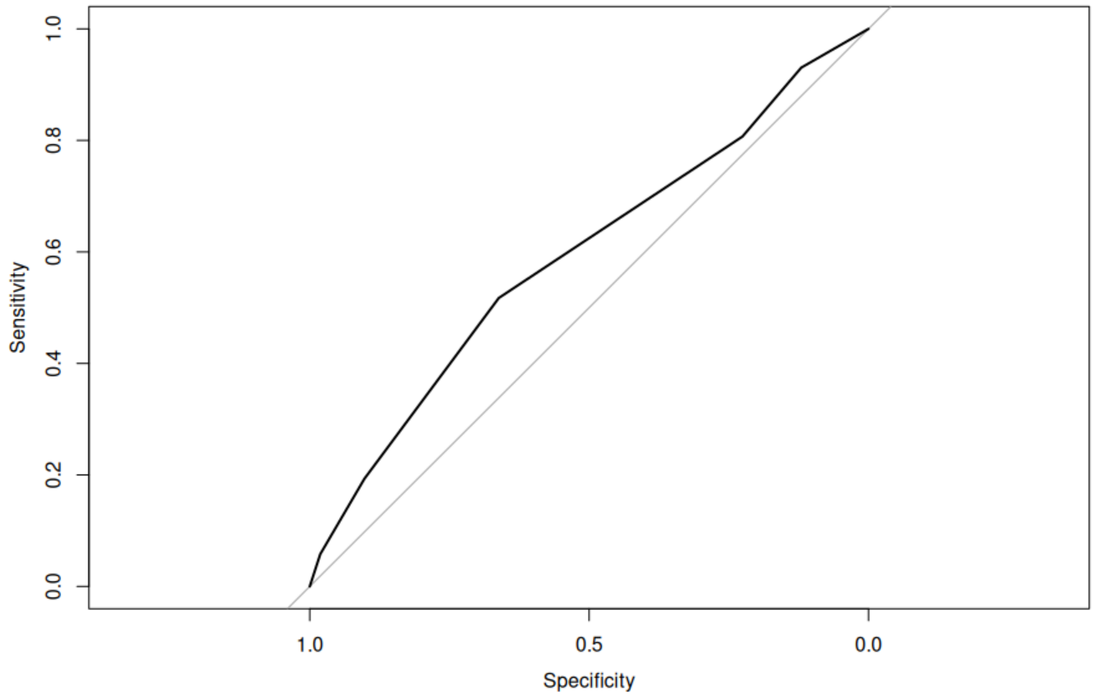

Figure B.7: ROC curve for the pruned classification tree